CHE-225 Introduction to Chemical Engineering Analysis **Spring 2023**

PROBLEM SET 1

(due Jan 20, 2023 @ 8:30 AM in Gradescope)

Problem 1 (20 pts total)

A liquid mixture containing 30.0 mole % of benzene (B), 25.0 mole % of toluene (T), and the balance of xylenes (X) is fed to a distillation column. The bottom product contains 98.0 mole % of X and no B; moreover, 96.0% of the X in the feed is recovered in this stream. The overhead product is fed to a second column. The overhead product from the second column contains 97.0% of the B in the feed to this column. The composition of this stream is 94 mole % of B and the balance of T. The second column has both overhead and bottom streams as well.

a) Draw and label a flowchart of this process and do the degree of freedom analysis to prove that for an assumed basis of calculation, molar flow rates and compositions of all process streams can be calculated from the information given. Write in order the equations you would use (and solve) to calculate unknown process variables. In each equation (or a pair of simultaneous equations) circle the variable(s) for which you would solve. Do NOT do the calculations.

b) Calculate: 1) the percentage of B in the process feed (*i.e.*, the feed to the first column) that emerges in the overhead product from the second column; and 2) the percentage of T in the process feed that emerges in the bottom product from the second column.

Problem 2 (35 pts total)

Butane is burned with air. No carbon monoxide is present in the combustion product.

a) Use a degree of freedom analysis to prove that if the percentage excess air and the percentage conversion of butane are specified, the molar composition of the product gas can be determined.

b) Calculate the molar composition of the product gas for each of the following three cases: 1) theoretical air supplied and 100% conversion of butane; 2) 20% excess air and 100% conversion of butane; and 3) 20% excess air and 90% conversion of butane. (Assume a basis of calculation)

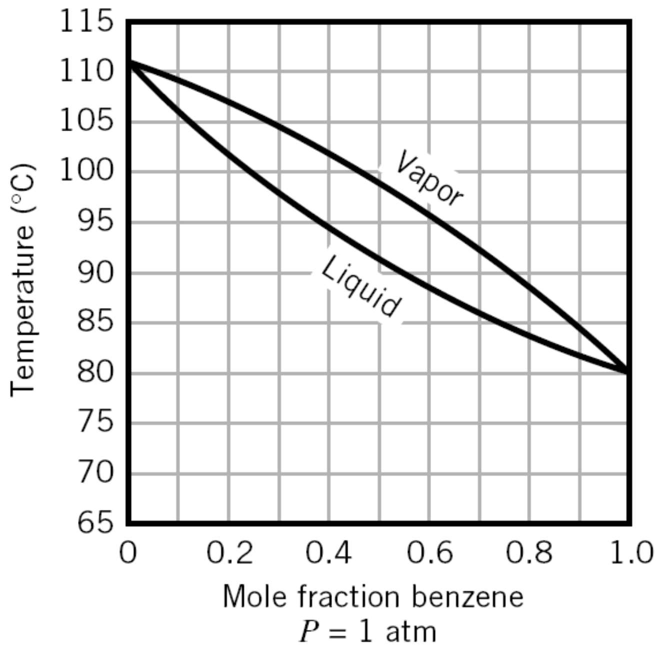
Problem 3 (15 pts total)

A vapor mixture containing 30 mole % of benzene (B) and 70 mole % of toluene (T) at 1 atm is cooled isobarically in a closed container from an initial temperature of 115°C. Use the Txy diagram (attached below) to answer the following questions:

a) At what temperature does the first drop of condensate form? What is its composition?

b) At one point during the process the system temperature is 100°C. Determine the mole fraction of B in the vapor and liquid phases, and the ratio of total moles in vapor/total moles in liquid at this point (assume a basis of calculation).

c) At what temperature does the last bubble of vapor condense? What is its composition?



Problem 4 (30 pts total)

Ethanol is produced commercially by the hydration of ethylene

$$C_2H_4(g) + H_2O(v) \leftrightarrow C_2H_5OH(v)$$

Some of the product is converted to dietyl ether (*i.e.*, $(C_2H_5)_2O$) in the undesired side reaction

$$2C_2H_5OH(g) \leftrightarrow (C_2H_5)_2O(v) + H_2O(v)$$

The combined feed to the rector contains 57.3 mole% C_2H_4, 36.7 mole% H_2O and the balance nitrogen. The feed enters the reactor at 310°C and the reactor operates isothermally at the same temperature as the input stream. An ethylene fractional conversion of 5% is achieved, and the yield of ethanol (*i.e.*, moles of ethanol produced/moles of ethylene consumed – see chemical reaction) is 0.90.

a) Calculate the reactor heating or cooling requirement in kJ/mol feed (Use Enthalpy table and assume a basis of calculation)

b) Why would the reactor be designed to yield such a low conversion of ethylene? What processing step(s) would probably follow the reactor in a commercial implementation of this process?

Note:

You can find thermodynamic data for ethylene, water, methanol and nitrogen in the various Appendices in F&R&B. Data for dietyl ether are as follows:

$\Delta\hat{H}_f^{\circ}$=-272.8 kJ/mol (for liquid diethyl ether)

$\Delta\hat{H}_v$ =26.05 kJ/mol (to the first approximation assume the same value at any temperature)

$c_P = 0.08945 + 40.33\text{x}10^{-5}\,\text{T} - 2.444\text{x}10^{-7}\,\text{T}^2$ (T is in °C and c_p is in kJ/mol/°C)